

SEVANEST

TECHNICAL REQUIREMENTS & DESIGN DOCUMENT

Verified Help. Trusted Homes.

Document Type	Technical Requirements & Design Document
Product	SevaNest — Household Workforce Marketplace
Platforms	Android, iOS & Web Administration Portal
Document Version	Version 1.0

Client-ready technical baseline

Contents

1. Document Purpose
2. Product & Solution Overview
3. Scope & Service Categories
4. User Roles & Access Model
5. Functional Requirements
6. Core Business Workflows
7. Technical Architecture
8. Data Model Requirements
9. API & Integration Requirements
10. Security & Privacy Requirements
11. Non-Functional Requirements
12. Reporting & Analytics
13. Configuration & Business Administration
14. Commercial & Payment Capability
15. Deployment & Operations
16. Quality Assurance & Acceptance
17. Operational & Audit Controls
18. Future-Ready Architecture
19. Technical Deliverables
20. Solution Summary

1. Document Purpose

This document defines the technical requirements, solution architecture, application behavior, security controls, data requirements, integrations, and operational capabilities for SevaNest. It is intended to provide a clear technical baseline for implementation, validation, deployment, and ongoing operation of the platform.

- Define the functional and technical behavior of the Customer App, Worker App, and Web Administration Portal.
- Establish a scalable architecture for a household workforce marketplace.
- Define security, privacy, verification, payment, notification, reporting, and audit requirements.
- Provide implementation-level guidance while keeping business rules configurable through administration.

2. Product & Solution Overview

SevaNest is a household workforce marketplace connecting customers with verified maids, drivers, cooks, babysitters, elderly-care helpers, and cleaning workers. The platform supports recurring household employment and short-term service requirements.

Component	Purpose	Primary Users
Customer Mobile App	Discover workers, request interviews/trials, confirm bookings, make payments, track services, rate workers, and raise support/replacement requests.	Customers
Worker Mobile App	Register, complete profile and KYC, manage skills and availability, receive opportunities, manage interviews/jobs, track attendance/status, and raise support requests.	Workers
Web Administration Portal	Operate the marketplace, verify users/documents, manage bookings, payments, support, configuration, reporting, and audit activities.	Operations, Verification, Support, Super Admin
Backend & API Layer	Central business logic, authentication, matching, booking, payments, notifications, support, reporting, and integrations.	All applications

3. Scope & Service Categories

The platform shall support the following service categories. Categories and related configuration shall be managed as administrative masters so additional services can be introduced without redesigning the applications.

Category	Supported Engagements	Typical Use
House Maid	Part-time, Full-time, Live-in	Cleaning, utensils, household support
Driver	Hourly, Daily, Monthly	Personal/family or temporary driving
Cook	Part-time, Monthly	Meal preparation
Babysitter	Hourly, Full-time	Child supervision and basic care
Elderly Care Helper	Shift, Full-time	Non-clinical assistance and companionship
Cleaning Helper	One-time, Hourly	Basic or deep cleaning assistance

4. User Roles & Access Model

Role	Core Responsibilities
Customer	Create household requirements, discover workers, shortlist, interview/trial, book, pay, track service, rate, request replacement, and raise support tickets.
Worker	Create work profile, submit verification information, define skills/availability, receive opportunities, accept/decline, attend interview/trial, manage current job, and raise support requests.
Operations Admin	Manage customers/workers, matching, bookings, assignments, replacements, operational monitoring, and marketplace configuration.
Verification Executive	Review KYC/background information, approve/reject verification items, maintain remarks and re-verification status.

Role	Core Responsibilities
Support Executive	Manage support tickets, complaints, disputes, escalations, and resolution tracking.
Super Admin	Manage platform configuration, roles, policies, reports, security-sensitive operations, and audit access.

All administrative access shall be role-based. Sensitive documents and security-critical actions shall be restricted to explicitly authorized roles.

5. Functional Requirements

5.1 Customer Application

- Mobile OTP authentication.
- Customer profile containing name, email, address, preferred language, and saved addresses.
- Basic customer verification.
- Service selection with engagement type, location, preferred timing, start date, and applicable salary/budget range.
- Worker discovery using location, availability, service category, and relevant preferences.
- Worker cards showing profile photo, experience, skills, language, area, availability, verification status, ratings, and review summary.
- Shortlist/favourite workers.
- Interview or trial request with preferred date and time.
- Worker confirmation and booking status timeline.
- Payment through supported gateway methods including UPI, cards, and net banking.
- Receipts, transaction history, and refund status where applicable.
- Attendance/service status visibility, service history, ratings, support requests, and replacement requests.

5.2 Worker Application

- Mobile OTP registration.
- Profile photo, address, emergency contact, and previous employer/reference details.
- KYC/government identity document upload.
- Selection of one or more roles, skills, experience, languages, preferred locations, expected salary, availability, and engagement preferences.
- Receive job opportunities and view customer requirement summaries.
- Accept or decline opportunities.
- Interview/trial scheduling, job confirmation, and start-date visibility.
- Current job details, attendance/status, earnings/payment reference where applicable, ratings, and support/complaint access.
- Simple mobile UX with large controls, limited text, clear status labels, short multi-step forms, and local-language readiness.

5.3 Administration Portal

Module	Required Capabilities
Dashboard	Bookings, active workers, pending verification, complaints, replacements, and collections.
Customer Management	Profiles, status, bookings, history, complaints, and operational notes.
Worker Management	Profiles, documents, availability, skills, ratings, blacklist/suspension status.
Verification	Document review, approval/rejection, status, expiry/recheck, and reviewer remarks.
Booking Management	Request, match, interview, trial, confirmation, cancellation, and replacement.
Payments	Transaction status, fee records, refunds, invoice/reference data, and exports.

Module	Required Capabilities
Support	Ticket creation, assignment, escalation, resolution, and SLA tracking.
Reports	Booking, conversion, category demand, area demand, worker utilization, and operational metrics.
Configuration	Categories, service areas, fees, plans, notification templates, and administrative roles.

5.4 Trust, Verification & Safety

- Workers shall not be presented as fully verified until configured verification checks are completed and approved by an authorized user.
- Identity verification shall support government ID/document upload and manual or approved partner verification.
- Address and previous-employment/reference information shall be captured and tracked.
- Police verification status/document tracking shall be supported where applicable.
- Customer mobile, address, and basic profile verification shall be supported.
- Report/block, complaint, escalation, suspension, and emergency-contact visibility controls shall be available according to configured policy.
- Full identity documents shall never be publicly exposed.
- Critical verification, payment, user-status, and administrative actions shall be auditable.

5.5 Booking & Matching

- Customer creates a service requirement containing category, engagement type, location, timing, start date, and applicable budget.
- System retrieves suitable workers based on configured service, location, availability, and profile criteria.
- Customer can shortlist a worker or request an interview/trial.
- Operations users may manage or override matching and assignment according to permissions.
- Booking shall progress through controlled states and retain a complete status history.
- Cancellation, confirmation, replacement, and support events shall be linked to the relevant booking/engagement.

5.6 Replacement Workflow

- Customer initiates a replacement request and provides a reason and supporting notes.
- Support/Operations reviews eligibility and updates engagement status.
- Alternative workers are identified using configured matching criteria.
- Customer selects the replacement where applicable.
- New worker confirmation is recorded without deleting historical engagement records.
- Replacement activity remains auditable.

5.7 Ratings, Reviews & Support

- Ratings shall be associated with completed or otherwise eligible engagements.
- Customers and workers shall be able to raise support/complaint requests.
- Support tickets shall have category, priority, owner, status, escalation, and resolution information.
- Complaint and resolution history shall remain available to authorized operational users.

6. Core Business Workflows

6.1 Customer Journey

- OTP login
- Select service
- Choose engagement type and timing
- Enter location

- View verified workers
- Shortlist or request interview
- Confirm worker
- Pay applicable fee
- Worker starts service
- Track, rate, or request replacement

6.2 Worker Journey

- OTP registration
- Complete profile
- Upload KYC/documents
- Set skills, locations, and availability
- Verification by authorized admin
- Receive job match
- Accept opportunity
- Attend interview/trial
- Job confirmation
- Start service and maintain status/attendance

6.3 Verification State Model

- Pending submission
- Under review
- Additional information required
- Approved/Verified
- Rejected
- Expired / Re-verification required

6.4 Booking State Model

- Requirement created
- Matching
- Interview requested
- Trial scheduled (where applicable)
- Confirmed
- Active
- Completed
- Cancelled
- Replacement requested
- Replaced

Exact state transitions shall be permission-controlled and recorded in the audit trail.

7. Technical Architecture

Layer	Recommended Technology / Approach	Responsibilities
Mobile	Flutter	Single maintainable codebase for Android and iOS.
Admin Web	React	Responsive operations and administration portal.

Layer	Recommended Technology / Approach	Responsibilities
Backend	Node.js/NestJS or Java/Spring Boot	REST APIs, business rules, authorization, integrations, workflows.
Database	PostgreSQL	Transactional relational data and integrity constraints.
Cache / Queue	Redis and/or managed queue	OTP, caching, asynchronous jobs, notifications, and integration processing.
Cloud	AWS or Azure	Compute, networking, managed services, storage, monitoring, and backups.
Object Storage	Cloud object storage	KYC documents, profile media, and controlled file access.
Authentication	OTP + token-based sessions	Customer/worker login and secure session management.
Push Notifications	Firebase Cloud Messaging	Android/iOS push notifications.
Payments	Razorpay / Cashfree / PayU	Indian payment methods and transaction processing.
Maps	Google Maps or equivalent provider	Location, distance, and service-area functions.
Monitoring	Cloud logs, metrics, and alerts	Operational visibility and production issue detection.

7.1 Logical Components

- API Gateway / Application API
- Authentication & Session Management
- User/Profile Management
- Worker Verification
- Service Category & Availability
- Search & Matching
- Booking & Interview/Trial
- Payment
- Notification
- Support & Complaints
- Reporting & Administration
- Audit & Security

8. Data Model Requirements

Entity	Key Information
Customer	Profile, addresses, verification status, bookings, payments, ratings.
Worker	Profile, roles/skills, locations, availability, verification, jobs, ratings.
Service Category	Category, engagement types, status, configurable attributes.
Booking	Customer requirement, selected worker, dates, status, fee/payment references.
Interview / Trial	Schedule, status, outcome, notes.
Verification Document	Document type, secure reference, status, reviewer, remarks, expiry/recheck.
Payment	Amount, fee type, gateway reference, status, refund details.
Attendance / Service Status	Date, status/check, exceptions, notes.
Replacement	Reason, eligibility/approval, original engagement, replacement engagement.
Support Ticket	Category, priority, owner, status, escalation, resolution.
Notification	Channel, template, recipient, delivery status, timestamps.

Entity	Key Information
Audit Log	Actor, role, action, entity, record reference, timestamp, relevant metadata.

8.1 Data Integrity Requirements

- Use relational foreign-key relationships for transactional entities.
- Use unique constraints for identities and other business identifiers where applicable.
- Use controlled status values rather than free-text workflow states.
- Store created_at and updated_at timestamps on persistent business entities; retain created/updated actor information where auditability requires it.
- Use soft-deactivation/status fields where historical records must remain auditable.
- Sensitive documents shall be stored outside primary transactional tables with secure object references and access controls.
- Payment and booking records shall maintain immutable gateway/reference identifiers.

9. API & Integration Requirements

Integration	Technical Requirement
Authentication / OTP	OTP generation, verification, expiry, retry/rate limiting, and secure token/session issuance.
Payment Gateway	Create payment, verify callback/webhook, update transaction state, receipt/reference, refund status.
Maps	Geocoding/location lookup, distance/service-area calculations as required.
Push Notifications	Token registration, template-based events, delivery status where supported.
SMS / WhatsApp	Configurable transactional notifications with provider abstraction.
Verification Provider	Optional partner integration; platform must retain a manual verification workflow as fallback.
Email	Transactional communication where configured for receipts, support, or operational notices.

9.1 API Standards

- Versioned REST APIs with consistent request/response structures.
- Authentication and authorization enforced server-side.
- Validation for all client-supplied data.
- Standardized HTTP status codes and machine-readable error responses.
- Idempotency for payment and other retry-sensitive operations.
- Pagination, filtering, and sorting for list endpoints.
- Correlation/request identifiers for troubleshooting.
- Audit logging for security-sensitive and administrative actions.
- Provider-specific integrations shall be isolated behind service interfaces to support replacement of external vendors.

10. Security & Privacy Requirements

- HTTPS/TLS for all application and API traffic.
- Encryption of sensitive data at rest where appropriate.
- Role-based access control for administration.
- Least-privilege access to KYC and sensitive records.
- Mask sensitive information in routine screens.
- Secure storage and rotation of application secrets and integration credentials.

- Rate limiting and abuse protection for OTP and API endpoints.
- Audit logs for verification, payment, booking-status, user-status, and privileged administrative changes.
- Automated database backups with protected access.
- Production monitoring, error alerts, and operational logging.
- Data retention and deletion rules shall be configurable according to the final approved legal/privacy policy.
- Terms, privacy policy, consent, and applicable verification processes shall be reviewed before production use.

11. Non-Functional Requirements

Area	Requirement
Performance	Common application screens should remain responsive under normal supported mobile network conditions.
Search	Search results should be returned efficiently with pagination and appropriate indexing.
Scalability	Architecture shall support expansion across additional service areas and cities without redesigning core modules.
Reliability	External integration failures shall be handled with retries, timeouts, clear failure states, and safe reconciliation.
Availability	Production infrastructure shall use appropriate managed services, health monitoring, backups, and recovery procedures.
Maintainability	Business configuration such as categories, fees, areas, templates, and roles should be admin-managed where practical.
Accessibility / Usability	Worker workflows shall prioritize large controls, clear labels, short forms, and local-language readiness.
Auditability	Critical business and administrative events shall retain actor, action, record, and timestamp information.

12. Reporting & Analytics

- New customers by day/week/month.
- Worker registrations and verification conversion.
- Booking requests versus confirmed bookings.
- Category-wise and area-wise demand.
- Worker utilization.
- Interview-to-confirmation conversion.
- Cancellation and replacement rates.
- Payment collections and transaction status.
- Support ticket volume, priority, and resolution time.
- Top-rated workers and repeat customers.

13. Configuration & Business Administration

- Service categories and engagement types.
- Serviceable areas and operational coverage.
- Booking/platform fees and configurable revenue rules.
- Subscription/plan configuration where enabled.
- Notification templates and communication channels.
- Administrative roles and permissions.
- Verification rules, statuses, and re-verification settings.
- Support categories, priorities, and escalation rules.

- Replacement eligibility/configuration.
- Operational status values and business rules.

Fee values and commercial rules shall be configurable through authorized administration rather than hard-coded in mobile applications.

14. Commercial & Payment Capability

- Support configurable booking fees.
- Support configurable placement fees.
- Support customer subscription plans where enabled.
- Support optional verification fees.
- Support replacement-plan coverage where enabled.
- Support apartment/community partnership plans where applicable.
- Support employer/bulk workforce services where applicable.
- Payment records must reconcile gateway status with the internal booking and transaction state.

15. Deployment & Operations

- Separate development, testing/staging, and production environments.
- Source-controlled application and infrastructure configuration.
- Automated build and deployment pipelines where supported.
- Secure environment-specific secrets and credentials.
- Database migration management.
- Centralized application and infrastructure logging.
- Monitoring and alerting for API failures, background jobs, payment callbacks, and critical infrastructure health.
- Scheduled database backups and tested restoration procedures.
- Controlled production access with role-based permissions.
- Operational documentation covering deployment, configuration, support, and incident handling.

16. Quality Assurance & Acceptance

- Functional testing across Customer, Worker, and Admin workflows.
- API and integration testing for OTP, payment, notifications, maps, and verification workflows.
- Regression testing for booking, payment, replacement, and support state transitions.
- Android/iOS device coverage and supported web-browser testing.
- Security testing for authentication, authorization, sensitive-data access, and common API abuse scenarios.
- Performance testing for search and critical transactional APIs.
- User acceptance testing against agreed acceptance criteria.
- Critical production workflows must operate without unauthorized state transitions or loss of audit history.

16.1 Core Acceptance Scenarios

- Customer can register through OTP and create a valid profile.
- Worker can register, complete work preferences, upload required documents, and submit verification.
- Authorized administrator can approve/reject verification items with remarks.
- Customer can search and filter suitable workers and view verification/rating information.
- Customer can request an interview/trial and complete a booking.
- Payment success/failure is correctly reflected against the relevant transaction and booking.

- Authorized administrators can manage permitted booking states.
- Customer can raise replacement and support requests.
- Ratings are restricted to eligible engagements.
- Unauthorized administrative users cannot access sensitive screens or KYC documents.

17. Operational & Audit Controls

- Every privileged action shall identify the acting user and timestamp.
- Verification changes shall preserve previous status and reviewer remarks.
- Booking status changes shall retain a history of transitions.
- Payment reconciliation events shall be traceable to gateway references.
- User suspension/activation actions shall be auditable.
- Access to sensitive documents shall be role-restricted and logged where technically feasible.
- Operational notes shall be associated with the relevant customer, worker, booking, or ticket.

18. Future-Ready Architecture

The architecture shall allow future capabilities to be introduced without redesigning the core platform. Potential extensions include multilingual interfaces, advanced worker matching, GPS/geofenced attendance, worker training and certification, salary/payroll assistance, insurance/benefit partnerships, apartment/community dashboards, corporate workforce requests, referral and loyalty programs, masked communication, advanced fraud analytics, and multi-city operations.

These capabilities are architectural considerations and are not required for the baseline release unless separately enabled.

19. Technical Deliverables

- Customer mobile application.
- Worker mobile application.
- Web administration portal.
- Backend/API services.
- Database schema and migration scripts.
- Cloud infrastructure and deployment configuration.
- API and integration documentation.
- Administrative user guide.
- Operational workflow documentation.
- Test summary and acceptance closure documentation.
- Production application builds and release assets.
- Knowledge-transfer and handover documentation.

20. Solution Summary

SevaNest shall be implemented as a secure, modular, API-driven marketplace platform with separate experiences for customers, workers, and operational administrators. The solution is centered on verified workforce discovery, controlled booking workflows, transparent payment and service status, operational support, and auditable administration.

The architecture should favor configurable business rules, strong data integrity, least-privilege access, provider abstraction, and scalable cloud services so that the platform can expand from its initial service categories and operating areas without requiring a fundamental redesign.